

Mixture of Specialized Experts for Model-Heterogeneous Personalized Federated Learning

Tonghui Liang[✉], *Student Member, IEEE*, Meng Hu, *Student Member, IEEE*,
and Enchang Sun[✉], *Senior Member, IEEE*

Abstract—Federated learning is a distributed machine learning paradigm designed to facilitate collaborative model training while preserving user data privacy. However, in practical scenarios, data and model heterogeneity have emerged as significant barriers to its widespread application. Existing model heterogeneous personalized federated learning (MHPFL) algorithms often face trade-offs between performance and cost of computing and communication. To address this, we propose FedMoEKD, a novel MHPFL algorithm based on the Mixture of Experts (MoE) framework with a dynamic routing mechanism and decoupled training strategy. This approach decouples local MoE training and enhances the specialization capabilities of local expert models through knowledge distillation. By focusing on expert specialization, it achieves superior performance compared to state-of-the-art methods while maintaining minimal computational and communication overheads, including at least 0.58% higher accuracy on CIFAR-10 and 58% lower computation overheads than FedProto. Extensive experiments conducted on multiple datasets demonstrate that FedMoEKD outperforms existing MHPFL approaches in both convergence speed and model accuracy.

Index Terms—Federated learning, model heterogeneity, mixture of experts, knowledge distillation.

I. INTRODUCTION

FEDERATED Learning (FL) is a decentralized learning paradigm that enables collaborative model training without the need to share raw data, offering inherent advantages in privacy preservation. Despite its promising potential, FL has yet to achieve the smooth and stable training initially envisioned, largely due to the presence of heterogeneity among clients. This heterogeneity stems from disparities in local data distributions and the need for different model architectures across clients.

Data heterogeneity refers to inconsistencies in the data distributions across clients, typically manifesting as non-independent and identically distributed (Non-IID) data induces statistical heterogeneity, where each client's data follows a distinct probability distribution, directly degrading the

performance and convergence of federated training. **Model heterogeneity** arises when clients employ different local model architectures. Personalized models tailored to local Non-IID data often outperform a global expert model in such settings. Traditional FL methods assume a globally shared model structure among all clients. While prior research has largely addressed FL under heterogeneous data with homogeneous models, recent attention has shifted toward tackling Model Heterogeneous Personalized Federated Learning (MHPFL), where both data and model heterogeneity coexist among clients.

Current MHPFL methods can be broadly categorized into two types, based on whether they require a homogeneous model for global aggregation: **(1) Methods that aggregate heterogeneous models without relying on a shared homogeneous model or subcomponents.** FedKT [1] leverage knowledge distillation on a public dataset to align models with different architectures. However, it requires access to a public dataset whose distribution approximates that of the teacher model. In practice, such datasets are difficult to obtain, both for aggregating client-to-server knowledge and for disseminating server-to-client information. To address this limitation, FedGen [2] propose training a global generator that synthesizes data aligned with the distribution of the teacher model. Although this alleviates the dependency on real public datasets, training such generators incurs additional computational overhead. Alternatively, methods such as FedDistill [3], FedProto [4], and FedGH [5] bypass the need for public or synthetic datasets altogether. These approaches involve sharing local class information and corresponding average logits or feature representations with the server, which are then used for distillation against global representations. However, although these approaches do not require additional model structures, both knowledge distillation and generator training introduce substantial computational overhead. **Methods that rely on aggregating partially homogeneous models or components.** For instance, FedRep [6] decouple the local model into a shared homogeneous feature extractor and a personalized classification head, aggregating only the shared components. Similarly, LG-FedAvg [7] support partial heterogeneity, but still rely on the existence of shared substructures, which may constrain model expressiveness and fail to fully adapt to diverse model architectures. Other approaches, such as FML [8] and FedKD [9], assign each client a heterogeneous large model and a shared small model. These are trained via mutual knowledge exchange to transfer local and global information. Building upon this concept, pFedMoE [10] introduces a Mixture-of-Experts (MoE) structure where global

Received 30 March 2025; revised 17 June 2025; accepted 2 July 2025. Date of publication 7 July 2025; date of current version 14 October 2025. The associate editor coordinating the review of this article and approving it for publication was O. M. Caicedo Rendon. (*Corresponding author: Enchang Sun.*)

Tonghui Liang and Enchang Sun are with the School of Information Science and Technology, Beijing University of Technology, Beijing 100124, China (e-mail: tonghui1111@emails.bjut.edu.cn; ecsun@bjut.edu.cn).

Meng Hu is with the Department of Software Engineering, Zhejiang University, Hangzhou 310027, Zhejiang, China (e-mail: 22351081@zju.edu.cn).

Digital Object Identifier 10.1109/LNET.2025.3586347

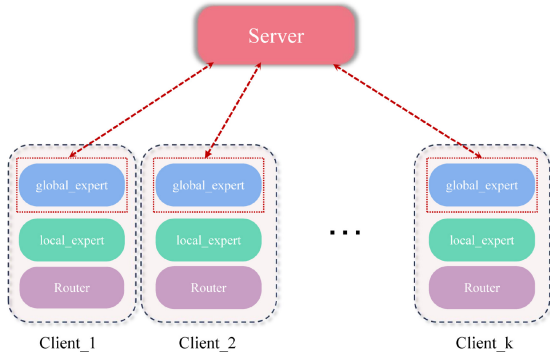


Fig. 1. Server aggregate homogeneous global expert model.

homogeneous and local heterogeneous models are decoupled and trained independently.

Nevertheless, existing MHPFL methods face critical limitations: (1) dependency on public/synthetic datasets or shared substructures, (2) substantial computational overhead from generator training or distillation, and (3) constrained expressiveness due to partial homogeneity requirements. These gaps motivate the design of modular architectures with decoupled expert specialization and efficient communication.

Inspired by these developments, FedMoEKD is proposed to better exploit the specialization potential of MoE experts by ensuring the effectiveness of expert modules within local MoE models. MoE divides the learning task among multiple expert models and uses a gating mechanism to selectively activate them based on the input. While address model heterogeneity by decoupling specialized local experts (for personalization) from shared global experts (for generalization) [10]. In FedMoEKD, each client is equipped with a shared homogeneous global model in addition to its local heterogeneous model, as shown in Fig. 1. A routing mechanism is employed to balance the personalization capacity of local experts with the generalization ability of global experts. To preserve the specialized performance of local experts, we decouple their training from the rest of the model and propose an innovative knowledge distillation method tailored to local client data, replacing conventional training methods. Drawing upon techniques from pFedKD [11] and Decoupled Knowledge Distillation [12], our approach significantly enhances local expert personalization. The effectiveness of our proposed algorithm is demonstrated through extensive experiments, which demonstrate that it surpasses state-of-the-art (SOTA) baselines in accuracy while maintaining high training efficiency. In summary, the main contributions of this letter are as follows:

- We propose a method inspired by the MoE framework that utilizes a routing mechanism during the local training phase in federated learning to adaptively balance the client-level generalization and personalization.
- We further enhance the specialization of expert models by introducing a decoupled training strategy for local MoE modules, enabling more effective training of both expert networks and the router.
- We further incorporate knowledge distillation to accelerate the convergence and improve the performance upper bound of local expert models.

The remainder of this letter is organized as follows: Section II details the FedMoEKD framework, Section III presents experimental results, and Section IV concludes with a discussion of future work. The source code of our experiments is now available at: <https://github.com/thuiaaa/MHPFL>.

II. MIXTURE OF SPECIALIZED EXPERTS FOR MODEL HETEROGENEOUS PERSONALIZED FEDERATED LEARNING

In FL, the global model possesses rich generalization knowledge, while local models contain personalized knowledge. Participating clients aim to leverage their limited local data to enhance the generalization capabilities of their local models, thereby improving overall model performance. Our objective is to complete the training of model-heterogeneous federated learning without increasing the communication and computational burden of FL. Specifically, we aim to achieve superior personalization performance on local data compared to the global model while maintaining an excellent generalization capability in the global model. To avoid exposing the structure of local models and reduce the computational burden on the server, we assign each client a small homogeneous expert model to provide global generalization capabilities and participate in the global training of federated learning.

For client k participating in training, a heterogeneous local expert model is deployed locally as $\mathcal{E}_{local,k}(w_k) = \mathcal{E}_{local,k}^{ex}(w_k) \circ \mathcal{E}_{local,k}^{head}(w_k)$, where $\mathcal{E}_{local,k}^{ex}(w_k)$ represents the part of the local model excluding the classification head $\mathcal{E}_{local,k}^{head}(w_k)$. The local expert model $\mathcal{E}_{local,k}$ is responsible for extracting personalized information from client k 's local data. Similarly, a homogeneous global expert model is defined as $\mathcal{E}_{global,k}(v_k) = \mathcal{E}_{global,k}^{ex}(v_k) \circ \mathcal{E}_{global,k}^{head}(v_k)$.

Compared to prior work, we argue that the essence of expert models lies in their specialized capabilities within specific domains: global expert models exhibit superior generalization capabilities, while local expert models demonstrate stronger personalization capabilities. In each round of local training in federated learning, we sequentially train the local expert model, the global expert model, and the router network. Recent studies on knowledge distillation indicate that a student model can inherit the capabilities of a teacher model on the corresponding distilled data. Furthermore, research on the role of teacher models in knowledge distillation [13] suggests that an appropriate teacher model can accelerate the training of the student model in the early stages and act as a regularizer to prevent overfitting in the later stages. This approach enhances the upper bound of the local expert model's performance on local data while mitigating overfitting, thereby improving the overall model's performance. In subsequent sections, we conduct experiments and ablation studies to analyze the impact on accuracy, training speed, and convergence.

Overall, our training process is as follows: During the global training phase of federated learning, the server and clients exchange only the weights of the global expert model $\mathcal{E}_{global,k}$ for weighted aggregation and updates. Since the global expert model is an independent, small, homogeneous model, the number of parameters to be transmitted is relatively small, and no local model structure is exposed. If necessary, further compression techniques such as quantization can be applied

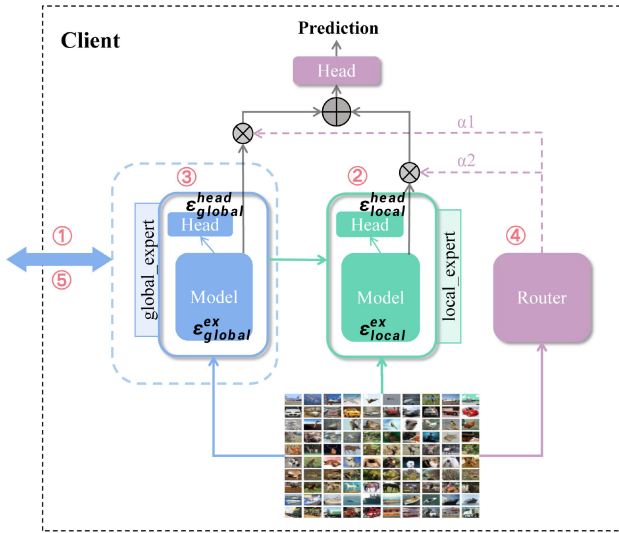


Fig. 2. Workflow of local update of FedMoEKD.

to the global expert model to further reduce data transmission requirements. We focus on the local update phase of each client, as illustrated in Fig. 2. For client k , if selected to participate in the t -th round of global training:

- (1) Client k receives the aggregated global model from the server, which is used as the local global expert model $\mathcal{E}_{global,k}(v_k) = \theta_{global}^{t-1}$.
- (2) On local data, knowledge distillation is performed using the global expert model $\mathcal{E}_{global,k}(v_k)$ to train the local expert model $\mathcal{E}_{local,k}(w_k)$.
- (3) The global expert model $\mathcal{E}_{global,k}(v_k)$ is updated on local data.
- (4) The parameters of $\mathcal{E}_{global,k}(v_k)$ and $\mathcal{E}_{local,k}(w_k)$ are frozen, then router network $\mathcal{G}_k$ is trained.
- (5) Upon completion of local training, the global expert model $\mathcal{E}_{global,k}$ is uploaded to the server for aggregation.

A. Knowledge Distillation for Training Heterogeneous Local Expert

To enhance the personalization capabilities of the local expert model, we employ knowledge distillation to train the student model. Based on prior research, we aim for the global expert model's capabilities on local data to guide the student model's training while acting as a regularizer in the later stages of training, thereby improving the local model's personalization performance. To compute the Kullback-Leibler (KL) divergence between the teacher and student models' outputs on local data, we retain the prediction heads of the expert models before updating the router network. During the t -th round of federated learning, client k performs knowledge distillation on its local data $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^N$, using the global expert model $\mathcal{E}_{global,k}$ to distill knowledge into the local expert model $\mathcal{E}_{local,k}$.

For each sample $(x_i, y_i) \in \mathcal{D}_k$, the input x_i is passed through the global expert model $\mathcal{E}_{global,k}(v_k) = \theta_{global}^{t-1}$ to generate a generalized representation:

$$z_{global,i} = \mathcal{E}_{global,k}^{ex}(x_i; v_k). \quad (1)$$

Similarly, the input x_i is passed through the local expert model $\mathcal{E}_{local,k}^{ex}(w_k)$ to generate a personalized representation:

$$z_{local,i} = \mathcal{E}_{local,k}^{ex}(x_i; w_k). \quad (2)$$

The heads of the global expert model $\mathcal{E}_{global,k}^{head}(v_k)$ and the local expert model $\mathcal{E}_{local,k}^{head}(w_k)$ are used to generate logits for the generalized and personalized representations, respectively. The knowledge distillation loss combines hard labels ($\mathcal{L}_{CE}$) and soft logits ($\mathcal{L}_{KL}$):

$$\begin{aligned} \mathcal{L}_{KD} = & (1 - \alpha) \mathcal{L}_{CE}(y_i, \mathcal{E}_{local,k}^{head}(z_{global,i})) \\ & + \alpha \mathcal{L}_{KL}(\mathcal{E}_{global,k}^{head}(z_{global,i}), \mathcal{E}_{local,k}^{head}(z_{local,i})) \end{aligned} \quad (3)$$

The local expert model is then updated via SGD: $w_k^t \leftarrow w_k^{t-1} - \eta_w \cdot \nabla \mathcal{L}_{KD}$.

To reduce the computational overhead, we set $\alpha = 0.5$ and $T = 1$ during updates. The knowledge distillation process focuses solely on aligning the output distribution of the local expert model with that of the global expert model to learn its capabilities on local data, without requiring any special parameter tuning to enhance performance. Although the teacher model may not contribute significantly at the beginning of training, as the local expert model rapidly improves through global training, knowledge distillation quickly yields efficiency gains and enhances the overall global performance. As global training converges, the global expert model stabilizes, providing a consistent teacher model for the local expert models across all clients.

After updating the heterogeneous local expert model, we update the global expert model on the local data:

$$\mathcal{L}_g = \mathcal{L}_{CE}(z_{global,i} \circ \mathcal{E}_{local,k}^{head}, y_i) \quad (4)$$

$$v_k^t \leftarrow v_k^{t-1} - \eta_v \cdot \nabla \mathcal{L}_g \quad (5)$$

B. Training Router Model

In large language models, Mixture-of-Experts (MoE) replaces feedforward neural networks with a combination of expert models, selecting which experts to activate through a routing mechanism. The router learns task-specific data patterns to dynamically allocate weights between experts, for example, assigning higher local expert weights to clients with highly Non-IID data, and higher global expert weights to clients with more representative samples. To enable direct feature fusion, we align the output dimensions of both expert models.

Local data is input into the router $R(\cdot)$, which determines the weights assigned to the two expert models and selects the proportion of activation for each. This routing decision is made through a lightweight MLP that learns to map input features to expert weight coefficients α_1 and α_2 . During training, after completing knowledge distillation for the local expert model and updating the global expert model, the outputs of $\mathcal{E}_{global,k}^{ex}(z_{global})$ and $\mathcal{E}_{local,k}^{ex}(z_{local})$ bypass their respective heads $\mathcal{E}_{global,k}^{head}$ and $\mathcal{E}_{local,k}^{head}$. Instead, they are weighted by the router's output coefficients α_1 and α_2 :

$$z_{combine,i} = \alpha_1 \cdot z_{global,i} + \alpha_2 \cdot z_{local,i} \quad (6)$$

TABLE I
TEST ACCURACY (%) ON CIFAR-10 AND CIFAR-100 WITH DIFFERENT PARTITIONING METHODS AND MODEL HOMOGENEITY SETTINGS

Partitioning	CIFAR-10				CIFAR-100			
	PAT		PRA		PAT		PRA	
Models	CNN_1	CNNs	CNN_1	CNNs	CNN_1	CNNs	CNN_1	CNNs
FedMoEKD	89.67	89.66	88.67	88.45	65.13	65.48	47.33	47.01
pFedMoE	79.38	79.2	81.78	81.9	61.72	61.61	42.29	42.26
FedProto	89.06	89.34	85.64	86.0	<u>65.15</u>	65.02	46.58	46.42
FedGen	88.57	88.86	87.83	88.12	61.76	61.65	42.25	42.31
FedDistill	88.99	88.93	87.65	87.64	64.24	64.11	44.67	44.5
FML	88.69	88.61	88.14	88.25	62.13	61.98	41.46	41.54
FedKD	88.91	88.7	88.08	88.16	64.51	64.29	46.34	46.32
LG-FedAvg	88.99	89.24	87.86	87.85	61.77	61.85	42.59	42.29
FedGH	89.09	89.35	85.14	84.45	61.89	61.81	41.79	41.54
FedTGP	89.09	89.46	85.86	85.13	30.72	25.36	47.25	<u>47.12</u>

The combined representation $z_{combine,i}$ is then passed through an aggregation head layer $\mathcal{F}_{combine}^{head}$ to train the router:

$$o_i = \mathcal{F}_{combine}^{head} \circ z_{combine,i} \quad (7)$$

The router is trained while keeping the parameters of the two expert models fixed, with local data at the client node. Under this decoupled training framework, FedMoEKD ensure that the homogeneous global expert model $\mathcal{E}_{global}(v_k)$ can be aligned with the heterogeneous local expert model by adding a linear layer to the $\mathcal{E}_{global}^{ex}$ component. This guarantees that both models have identical hidden-layer dimensions and router's update follows the loss function:

$$\mathcal{L}_{rt} = \mathcal{L}_{CE}(\mathcal{F}_{combine}^{head} \circ z_{combine,i}, y_i) \quad (8)$$

where $\mathcal{L}_{CE}$ represents the cross-entropy loss. Subsequently, all models are updated simultaneously via SGD optimizer.

III. EXPERIMENTS

A. Experimental Setup

To validate the effectiveness of our algorithm, we conducted extensive experiments on 4 NVIDIA GeForce RTX 4090 GPUs, comparing FedMoEKD with nine state-of-the-art (SOTA) algorithms on two benchmark datasets. For all methods, we used identical learning rates and performed ablation studies to verify the contributions of our proposed innovations.

Datasets: We evaluated FedMoEKD and other baselines on the CIFAR-10 and CIFAR-100 datasets. Both consist of 32×32 color images designed for classification tasks. CIFAR-10 contains 60,000 color images across 10 classes, with each class containing 6,000 images. CIFAR-100 comprises 100 classes, compared to CIFAR-10, CIFAR-100 features more fine-grained and diverse categories, with fewer samples per class.

Model Heterogeneity and Statistical Data Heterogeneity: In our experiments, we simulated both homogeneous and heterogeneous local models among clients. We prepared five distinct CNN models for participating clients. For the homogeneous global expert model, we uniformly adopted the smallest of the five CNN models. To simulate statistical data heterogeneity, we employed two widely-used partitioning strategies: Pathological and Practical. In the pathological heterogeneity setting, we assigned two disjoint classes from the CIFAR-10 and CIFAR-100 datasets to each client. In the practical

heterogeneity setting, the data distribution was controlled by a Dirichlet distribution, denoted as $Dir(\beta)$. There is a positive correlation between environmental heterogeneity and the value of β . For the default configuration, we set $\beta = 0.1$.

Comparison Methods: We compared FedMoEKD with nine MHPFL algorithms from the SOTA model, categorized according to the taxonomy outlined in Section I. MHPFL without Homogeneous Models: FedDistill [3], FedProto [4], LG-FedAvg [7], FedGen [2], FedGH [5]. MHPFL with Homogeneous Models: FML [8], FedKD [9], FedTGP [14], pFedMoE [10], and our proposed method, FedMoEKD.

Training Settings: To ensure fairness in experiments, we configured identical hyperparameters for all comparative methods. The learning rate was set to 0.01, and the batch size was fixed at 128. We implemented an autobreak mechanism to terminate training early if convergence criteria were met. We evaluated different numbers of local training epochs, varying numbers of participating clients, and different joining rates. The experimental results demonstrate that our method outperforms all other comparative methods across nearly all configurations.

B. Results and Discussion

We conducted extensive experiments on both datasets using two partitioning methods, Pathological (PAT) and Practical (PRA), under both homogeneous and heterogeneous model settings. To appropriately control the scale of the experiments, we performed comparative analyses across varying client participation rates in each round and different numbers of local training epochs.

1) *Performance Comparison:* On CIFAR-10, for the Pathological partitioning method, FedMoEKD outperforms the best-competing method by 0.58% in the homogeneous model setting and by 0.20% in the heterogeneous model setting. For the Practical partitioning method, our algorithm achieves improvements of 0.53% and 0.20% over the best-competing method in the homogeneous and heterogeneous model settings, respectively. On CIFAR-100, FedMoEKD consistently outperforms existing comparative approaches, achieving performance on par with or superior to FedProto and FedTGP, despite their computational and communication costs being several times higher. The results are presented in Table I.

2) *Additional Experiments:* To evaluate the effectiveness of our method under varying conditions, we conducted

TABLE II
CLIENT PARTICIPATION AND JOINING RATE

Model	CNNs		
	Clients (% Join)	10 (100%)	20 (50%) 50 (20%)
FedMoEKD		89.67	88.09 83.15
pFedMoE		79.38	75.78 71.23
FedProto		89.06	87.67 84.49
FedGen		88.57	87.11 82.93
FedDistill		88.99	87.13 80.71
FML		88.69	87.01 81.96
FedKD		88.91	87.27 83.98
LG-FedAvg		88.99	87.18 82.83
FedGH		89.09	87.33 82.91
FedTGP		89.09	86.75 84.12

TABLE III
PER-ROUND TRAINING TIME (S)

Method	Time (s)
FedMoEKD	5.82
pFedMoE	4.76
FedProto	13.91
FedGen	5.86
FedDistill	6.94
FML	5.57
FedKD	26.99
LG-FedAvg	8.92
FedGH	9.46
FedTGP	16.48

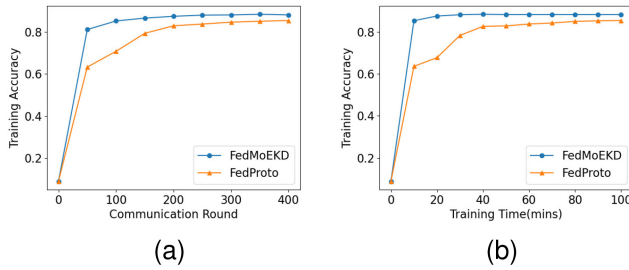


Fig. 3. Training efficiency compare with FedProto. (a) accuracy with communication round. (b) accuracy with training time.

experiments with different numbers of local training epochs $local_epoch \in \{1, 10\}$ and different levels of client participation during each epoch, corresponding to participation rates of $\{10 \text{ clients, } 100\%; 20 \text{ clients, } 50\%; 50 \text{ clients, } 10\%\}$. The results are presented in Table II.

3) *Training Efficiency*: We compared the per-round computation time of our algorithm with other SOTA baseline methods, presented in Table III. Additionally, we conducted a comparative analysis of the change in accuracy over time and training epochs between our algorithm and FedProto, the best-performing method among the baselines. The experimental results presented in Fig. 3 demonstrate that our algorithm exhibits excellent convergence speed and computational efficiency while maintaining competitive performance.

4) *Ablation Experiments*: To validate the effectiveness of the proposed method, we conduct an ablation study by comparing the following training strategies: joint training of expert models and the routing network (79.38%); sequential training where expert models are trained before the router (88.96%); fully decoupled training of all expert models and the router (89.13%); and an additional strategy incorporating knowledge distillation for local experts (89.66%).

The experimental results demonstrate that both the decoupled training of expert models and the application of knowledge distillation to local experts contribute positively to model performance, confirming the effectiveness of our design choices.

IV. CONCLUSION

In this letter, we propose FedMoEKD, a Mixture-of-Experts-based Model Heterogeneous Personalized Federated Learning algorithm to address both data and model heterogeneity in federated learning. By decoupling the training of local models and leveraging knowledge distillation to optimize local expert models, we enhance the specialization capabilities of expert components. Our method achieves performance that surpasses state-of-the-art baselines while maintaining low computational and communication overheads. Extensive experiments on multiple datasets demonstrate the feasibility and effectiveness of the proposed approach. In future work, we aim to further refine the routing mechanism in a more sophisticated manner, enhancing the robustness of the method and reducing the cost associated with model deployment.

REFERENCES

- [1] Q. Li, B. He, and D. Song, "Practical one-shot federated learning for cross-silo setting," 2020, *arXiv:2010.01017*.
- [2] Z. Zhu, J. Hong, and J. Zhou, "Data-free knowledge distillation for heterogeneous federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 12878–12889.
- [3] E. Jeong, S. Oh, H. Kim, J. Park, M. Bennis, and S.-L. Kim, "Communication-efficient on-device machine learning: Federated distillation and augmentation under non-IID private data," 2018, *arXiv:1811.11479*.
- [4] Y. Tan et al., "FedProto: Federated prototype learning across heterogeneous clients," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 8432–8440.
- [5] L. Yi, G. Wang, X. Liu, Z. Shi, and H. Yu, "FedGH: Heterogeneous federated learning with generalized global header," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 8686–8696.
- [6] M. A. Husnoo, A. Anwar, N. Hosseinzadeh, S. N. Islam, A. N. Mahmood, and R. Doss, "FedREP: Towards horizontal federated load forecasting for retail energy providers," in *Proc. IEEE PES 14th Asia-Pacific Power Energy Eng. Conf. (APPEEC)*, 2022, pp. 1–6.
- [7] P. P. Liang et al., "Think locally, act globally: Federated learning with local and global representations," 2020, *arXiv:2001.01523*.
- [8] T. Shen et al., "Federated mutual learning," 2020, *arXiv:2006.16765*.
- [9] C. Wu, F. Wu, L. Lyu, Y. Huang, and X. Xie, "Communication-efficient federated learning via knowledge distillation," *Nat. Commun.*, vol. 13, no. 1, p. 2032, 2022.
- [10] L. Yi et al., "pFedMoE: Data-level personalization with mixture of experts for model-heterogeneous personalized federated learning," 2024, *arXiv:2402.01350*.
- [11] T. Liang and E. Sun, "pFedKD: Personalized federated learning meet knowledge distillation," in *Proc. 6th Int. Conf. Elect., Control Instrum. Eng. (ICECIE)*, 2024, pp. 1–6.
- [12] B. Zhao, Q. Cui, R. Song, Y. Qiu, and J. Liang, "Decoupled knowledge distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11953–11962.
- [13] L. Yuan, F. E. Tay, G. Li, T. Wang, and J. Feng, "Revisiting knowledge distillation via label smoothing regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 3903–3911.
- [14] J. Zhang, Y. Liu, Y. Hua, and J. Cao, "FedTGP: Trainable global prototypes with adaptive-margin-enhanced contrastive learning for data and model heterogeneity in federated learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 16768–16776.